Reverse Engineering an Undocumented RF Garage Door Opener: A Passive Reconnaissance Study using SDR

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***Abstract*—This paper presents the passive reconnaissance and reverse engineering of a Remocom RMC-575 garage door remote. Initially hypothesized to utilize a secure rolling code protocol, hardware inspection revealed an undocumented 5751c cloned integrated circuit. By employing a Software Defined Radio (SDR V4), we successfully isolated the target transmission at 300.8 MHz, discarding environmental RF interference. The captured signal was demodulated and analyzed using Universal Radio Hacker (URH), requiring a precise symbol alignment of 470 samples/symbol. The extraction of a static hexadecimal payload definitively disproved the initial hypothesis, exposing the system's reliance on a highly vulnerable fixed code architecture susceptible to eavesdropping and cloning attacks.**

I. Introduction and Threat Model

This study focuses on the analysis of the Remocom RMC-575, a radio frequency remote control for a garage door. The primary objective of this project was to intercept the remote's signal, discover its operating frequency, and, if possible, successfully decode the transmitted data payload.

Our initial hypothesis was that, being a modern garage door opener, the device would likely employ a secure Rolling Code system. Such systems require dynamic synchronization with the receiver to prevent interception, similar to the encryption mechanisms used in modern automotive smart keys.

The current scope of this project is strictly limited to passive reconnaissance and signal analysis. It does not involve active transmission or the execution of replay attacks, although such active vulnerability testing may be considered for future work.

II. Hardware Reconnaissance

The first step in determining the operating frequency involved physically opening the remote control to inspect the printed circuit board (PCB) for any relevant hardware clues. During this inspection, the main integrated circuit was identified as a 5751c chip.

Further research revealed that this component is a Taiwanese clone lacking an official or comprehensive datasheet. The only actionable information recovered from secondary sources was a theoretical operating band spanning from 200 MHz to 870 MHz. Given that this frequency range is excessively broad for manual testing, it became necessary to employ a method to actively scan the radio frequency (RF) spectrum to pinpoint the exact transmission frequency.

III. Signal Acquisition

To sweep the broad frequency range and identify meaningful transmissions, an SDR V4 (Software Defined Radio) receiver was deployed in conjunction with the SDR++ software. This setup allowed for real-time monitoring of the RF spectrum and provided the capability to record raw data. The receiver was configured for Amplitude Modulation (AM) with a minimum gain of 20 dB to clearly distinguish the signal peaks from the noise floor.

During the initial signal hunting phase, a strong energy burst was detected at 433.5 MHz. Although the recorded waveform exhibited a synchronization sequence and strongly resembled a remote control code, its appearance was highly inconsistent. It did not reliably correlate with the physical button presses. Given that the 433 MHz band is a heavily populated ISM (Industrial, Scientific, and Medical) band full of interfering devices, this signal was discarded as environmental noise or a mere coincidence.

Finally, a consistent and distinct pulse was isolated at exactly 300.8 MHz. This transmission synchronized perfectly with every single button press, confirming it as the true target signal. The raw RF data was then recorded and exported as a `.wav` file for further digital analysis using the Universal Radio Hacker (URH) software.

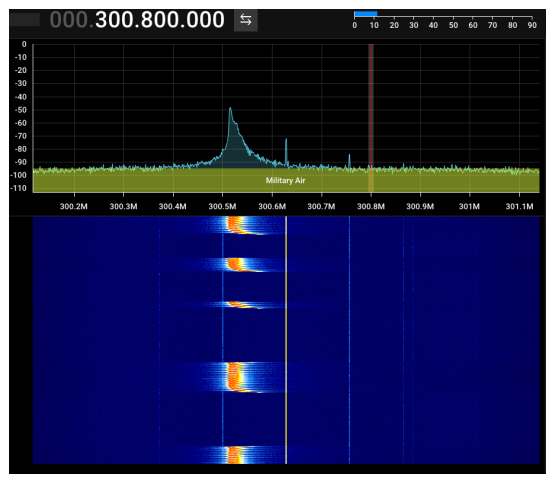

Fig. 1. Signal spectrum capture at 300.8 MHz using SDR++.

IV. Demodulation and Decoding

Upon importing the `.wav` file into Universal Radio Hacker (URH), we first visualized the raw captured analog signal. By switching the software's view to demodulated mode, the underlying binary digital waveform became clearly visible. The signal structure consisted of an initial high bit ('1') acting as a synchronization preamble, immediately followed by the data payload.

To ensure an accurate digital interpretation of the waveform, the symbol length parameter required manual calibration. By measuring the duration of a single '1' bit, the software was adjusted to exactly 470 samples per symbol. Following this crucial adjustment, the signal was perfectly decoded, revealing the static hexadecimal payload: `8080fefe8080fe...`

Furthermore, a temporal analysis of the capture revealed that the remote transmits the code burst every 57 milliseconds. Consequently, a standard human button press results in the entire payload being transmitted four or more times. While this redundant transmission is highly inefficient regarding battery power consumption, it significantly improves the system's reliability; if a single bit frame is cor-

rupted by environmental RF interference, the subsequent redundant frames ensure the garage door still opens reliably.

Regarding our initial threat model hypothesis, we observed that the decoded hexadecimal message remained entirely static across multiple physical button presses. This definitively proves that the remote does not utilize a secure Rolling Code mechanism. Instead, it relies on a legacy Fixed Code protocol, rendering the access control system highly vulnerable to eavesdropping, cloning, and replay attacks.

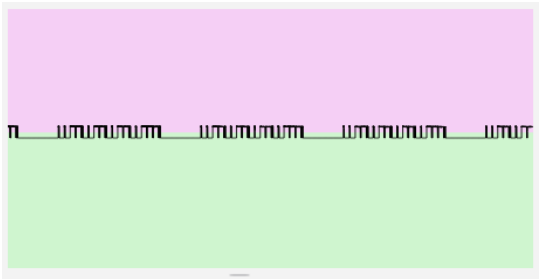

Fig. 2. Digital waveform demodulation and binary decoding process in URH, showing the 470 samples/symbol alignment.

V. Conclusion

In conclusion, this project successfully demonstrated the passive interception and decoding of an undocumented RF remote control. Despite the complete lack of official datasheets for the Taiwanese 5751c clone, the combination of SDR hardware and URH software proved highly effective for signal isolation and binary extraction.

Our analysis not only retrieved the underlying hexadecimal payload but also uncovered the redundant transmission mechanics of the protocol, explaining how the device balances energetic inefficiency with transmission reliability. Most importantly, the static nature of the decoded payload over multiple transmissions definitively refuted our initial rolling code hypothesis. This confirms that the Remocom RMC-575 relies on an insecure fixed code system, exposing the access control mechanism to trivial cloning and eavesdropping. Ultimately, this study highlights the critical security flaws inherent in legacy RF hardware and the power of software-defined radio for passive security auditing.

References

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